SECD2613 SYSTEM ANALYSIS AND DESIGN

SESSION 2025/2026 – SEMESTER 2

PHASE 2: INFORMATION SYSTEM GATHERING AND REQUIREMENT

FlexHub AI Marketplace [FlexServe]

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1.0 Overview of the Project

In this phase, questionnaires were used during this stage to extract information and requirements from the stakeholders. Such approaches were deemed necessary for a thorough understanding of user expectations and system needs, together with limitations of the current service discovery and booking mechanism.

A set of questionnaires was created and distributed through the online medium in order to obtain feedback from the stakeholders concerning the proposed FlexServe System. It comprised closed-ended questions only to provide answers that could be clearly identified within the context of user preference, common problems and expectations from the proposed system.

It is essential to note that the findings from the questionnaires play a crucial role in pinpointing some of the problems encountered by users such as difficulties in locating reliable service providers, real-time availability checks, lack of efficiency in communication, and absence of centralized service booking facility. Moreover, the findings were significant in providing insight into the functions desired in the proposed system like online booking, service search/filtering and appointment scheduling. The questionnaires were further analyzed by the stakeholders in terms of the accuracy and relevancy of the system requirements obtained.

The analysis of the AS-IS (as is) workflow of the booking service was performed based on the obtained information about the company's processes. The AS-IS workflow demonstrates that locating services and making bookings is done manually through various communication tools and social media channels. This often results in problems such as miscommunication, scheduling issues, difficulties in service discovery, and inadequate information record-keeping.

In order to support the system analysis, the workflow of the FlexServe System was presented in the form of a context diagram, also called a High-Level DFD (Data Flow Diagram). Next, a DFD that includes level 0 and its children was created in order to demonstrate data flows in greater detail. Such diagrams help understand how processes interact with each other and with data stores, with each process being able to be divided further into multiple subprocesses.

All of these steps together allow comprehending the system's requirements, as well as existing workflow problems and data flows. All of these aspects play a crucial role in the development of the proposed FlexServe System.

2.0 Problem Statement

The professional services industry faces many problems that make work slow and booking difficult for both customers and service providers. Based on the data from our questionnaire, the main problems are:

1. Hard to Find Customers and Services

- Service providers have to use their personal social media and friends to find customers. This makes it hard for them to get more business. At the same time, customers do not have one single website to easily search and find different service providers.

2. Slow and Messy Manual Booking

- All tasks, like keeping track of bookings and managing schedules are done by hand. Service providers have to reply to orders across many different apps, like WhatsApp or Telegram. This creates a lot of work and causes mistakes, like booking two customers at the same time slot (*double-booking*).

3. Slow Communication

- Customers and service providers only talk through normal chats and phone calls. This manual way is very slow. It takes a long time just to confirm booking details and fails to give quick answers to customer questions.

4. No Clear Reviews and Trust

- It is hard for customers to check prices, see ratings, or view past work portfolios before hiring someone. Because there is no proper review system, customers find it difficult to trust the quality and the price of the service.

5. Difficult Payment and Receipts

- Paying deposits or full payments is not convenient. Customers have to open separate bank apps to transfer money, take a screenshot of the receipt, and send it through chat. The service provider then has to check it one by one before they can confirm the booking.

These problems show that we need a good digital system to make scheduling automatic, help providers find more customers, and make payments easy. Fixing these issues is important to stop manual mistakes and build trust.

The proposed FlexServe system aims to solve these problems by creating easy online tools to modernize service discovery, make payments safe, and make the booking experience better for everyone.

3.0 Proposed Solutions

Booking errors, scheduling coordination problems and delays in customer communication are some of the operational inefficiencies caused by the current manual freelance service booking methods. We are trying to provide a new online service management system solution called FlexServe, which incorporates some of these automated technologies.

First, an automated scheduling and booking system will be created so that customers can finish the entire booking process, for instance from checking open dates to the final confirmation, all within the platform. The FlexServe service management system will handle availability verification, schedule blocking and booking requests automatically in a single interface. Customers can easily browse various service providers and hourly service prices with the system. By simplifying the booking process and eliminating the need for manual data entry, the FlexServe online service platform can help to improve customer satisfaction. Hence this can solve the problem of professional service providers struggling to record appointments manually using paper-written methods or scattered notebook diaries.

Next, an automatic notification system will be made available via FlexServe. Customers and service providers can receive updates and reminders immediately by using this notification system. Important details such as schedule modifications or any updates on the status of reservations will all be included in these alerts. By providing users with up-to-date information and maintaining communication, FlexServe can help prevent various problems such as the provider forgetting to send confirmation notifications or updates manually over text messages.

In addition, FlexServe will have a built-in communication interface. Customers can receive direct responses and support from the service providers to clarify specific job requirements before completing an appointment. Misunderstandings and response delays can be minimized by integrating a centralized communication option on the FlexServe platform. Additionally, by doing this, providers can free up their time to concentrate on service delivery rather than constantly checking text logs. As a result, miscommunication problems over disconnected platforms can be solved.

FlexServe will also provide a real-time update system which updates the profile comparison details of each independent service provider, including their availability schedules, active ratings and service fees in one place. As a result, the problem of repeated questions from customers due to not being able to find the provider’s operational details easily can be prevented. If the appointment timing or service details have any changes, this system will immediately update to block overlapping sessions. Thus, it can prevent scheduling conflicts like double-bookings and appointment overlaps from happening.

Furthermore, the FlexServe system will provide transaction processing functions like automatically calculating the total service fees and logging payment histories cleanly. Every independent service provider's hourly or session-based cost will be entered into the FlexServe system. This could improve the accuracy of determining the total amount that the customers must pay and can help to streamline transaction records. The automated tracking of total payment costs for each customer's order is possible in the FlexServe system by integrating a payment management module into the platform. This can help to prevent recording payment receipts wrongly or losing track of deposit data.

Additionally, FlexServe offers easy accessibility features and a user-friendly interface. The FlexServe platform will be made as user-friendly as possible so that the system can be used by people of all skill levels from all ages including customers and providers with limited computer skills. Users can quickly check open slots and verify costs using FlexServe which in return could improve the service's quality, ensuring that the system is usable by a larger audience, and raise user happiness and satisfaction.

FlexServe is dedicated to providing customers with a secure system hosting experience. In order to access the services provided and protect personal profiles, users must go through authentication and credential checks. Additionally, the system implements technological security controls and encryption mechanisms to safeguard private information. These security controls can help in protecting confidential transaction records, direct chat logs, personal details and financial transactions from unauthorized access.

4.0 Information gathering process

The information gathering phase is a critical component of the system development life cycle for the FlexServe platform. The primary objective of this phase is to extract information, capture, and evaluate the operational requirements of the current manual process and translate them into a fully automated, AI service management system. To build a high-fidelity requirement model, a digital questionnaire via Google Forms was made. This approach allowed the team to gather both qualitative and quantitative data systematically from a distributed base of stakeholders, consisting of service seekers and service experts. The deployed questionnaire comprised a total of 20 questions, balancing 8 closed-ended questions for quantitative metrics and 12 open-ended questions to capture rich, unconstrained operational feedback. The structural questions deployed across the survey, alongside the direct functional and non-functional requirements derived from the respondents feedback

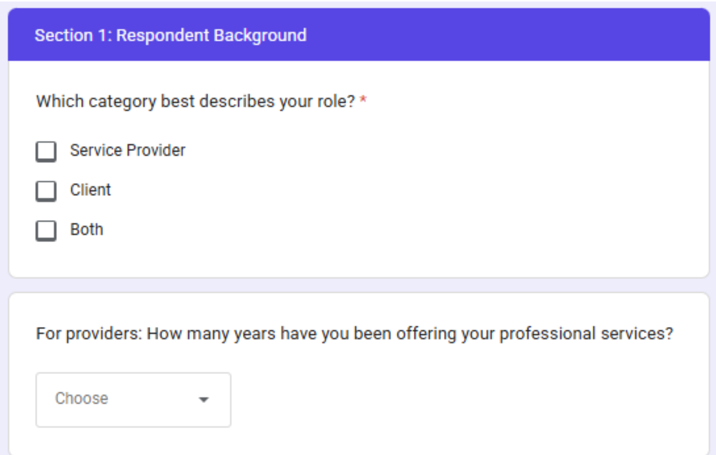
4.1 Method used

For the purpose of this phase of the project, the use of an extensive online survey through Google forms was applied as the extracting information technique of choice. This approach has helped the team collect structured requirements from a broad spectrum of fragmented sources including both client-side as well as expert-side users. The survey was further broken down into four main categories, each having a combination of closed as well as open-ended questions.

4.1.1 Questionaire

Section 1: Respondent Background

In this section, the profiles, booking frequency, and platforms that the targeted users are using are defined to understand the needs of the users.



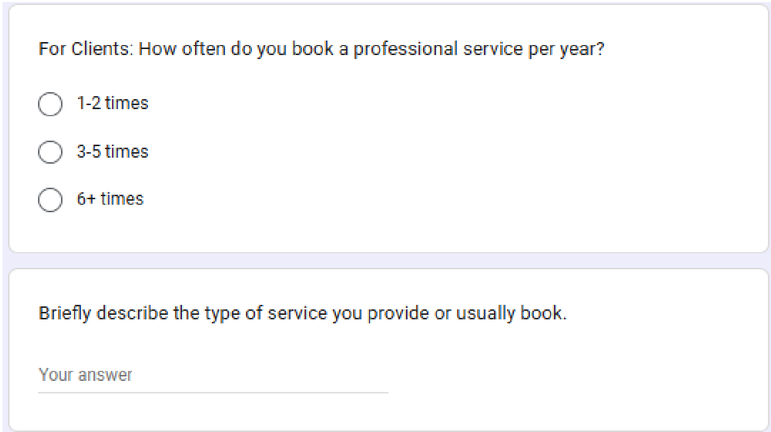
Section 1: Respondent Background

Which category best describes your role? *

- [ ] Service Provider
- [ ] Client
- [ ] Both

For providers: How many years have you been offering your professional services?

Choose



For Clients: How often do you book a professional service per year?

- 1-2 times
- 3-5 times
- 6+ times

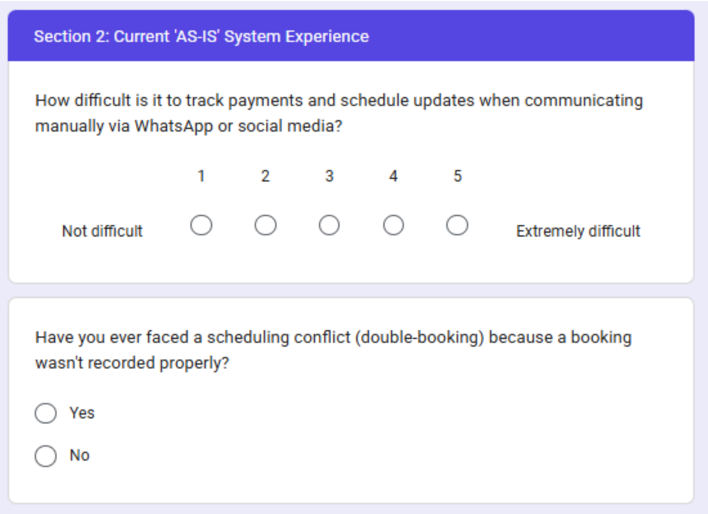
Briefly describe the type of service you provide or usually book.

Your answer

The respondents generally work in both capacities of both as user and service provider and depend a lot on WhatsApp, social media platforms like Instagram and Facebook, Google searches, and direct bank applications. Thus the design of the system should incorporate a Dual Role Interface within one account page and a one-stop solution for discovery, scheduling, and payment processes.

Section 2: Current System Experience (AS-IS)

This section has examined the user satisfaction aspect and highlighted some of the major operational issues facing the current manual booking system.

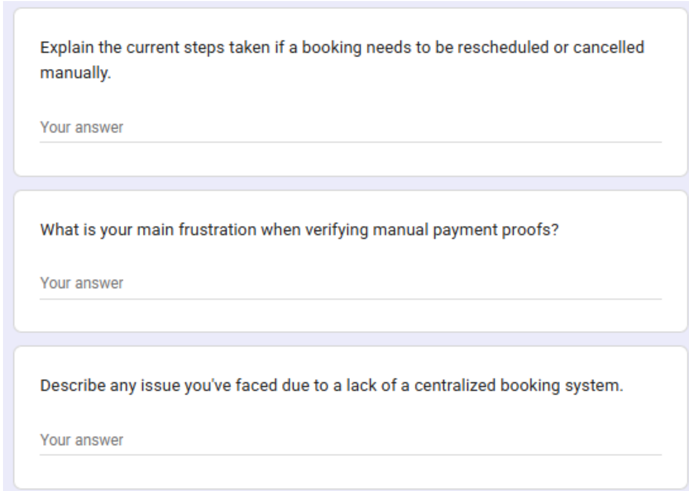

Section 2: Current 'AS-IS' System Experience

How difficult is it to track payments and schedule updates when communicating manually via WhatsApp or social media?

	1	2	3	4	5	
Not difficult	○	○	○	○	○	Extremely difficult

Have you ever faced a scheduling conflict (double-booking) because a booking wasn't recorded properly?

- Yes
- No


Explain the current steps taken if a booking needs to be rescheduled or cancelled manually.

Your answer

What is your main frustration when verifying manual payment proofs?

Your answer

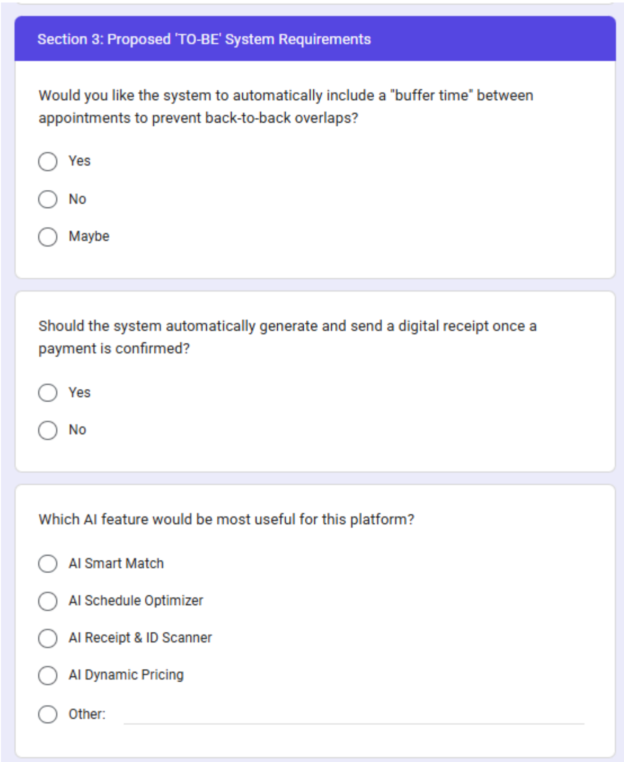
Describe any issue you've faced due to a lack of a centralized booking system.

Your answer

The feedback indicates that there was a high degree of dissatisfaction (rating 2 out of 5). It showed that the users faced serious friction due to the difficulty of going through the process of searching on Google for every single portfolio link, as well as coordinating via WhatsApp about their availability and avoiding double-booking and last-minute cancellations.

Section 3: Proposed System Requirements (TO-BE)

This section was dedicated to receiving direct feedback from users regarding their requirements in terms of automated functionalities and AI capabilities in the new platform.



Section 3: Proposed 'TO-BE' System Requirements

Would you like the system to automatically include a "buffer time" between appointments to prevent back-to-back overlaps?

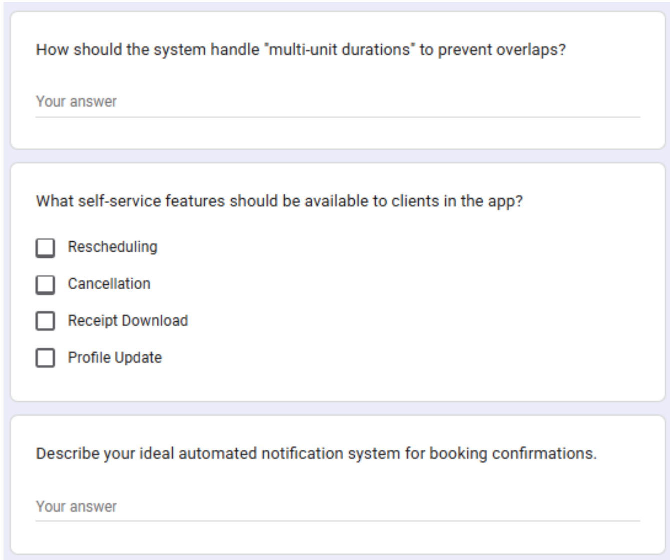
- Yes
- No
- Maybe

Should the system automatically generate and send a digital receipt once a payment is confirmed?

- Yes
- No

Which AI feature would be most useful for this platform?

- AI Smart Match
- AI Schedule Optimizer
- AI Receipt & ID Scanner
- AI Dynamic Pricing
- Other:



How should the system handle "multi-unit durations" to prevent overlaps?

Your answer

What self-service features should be available to clients in the app?

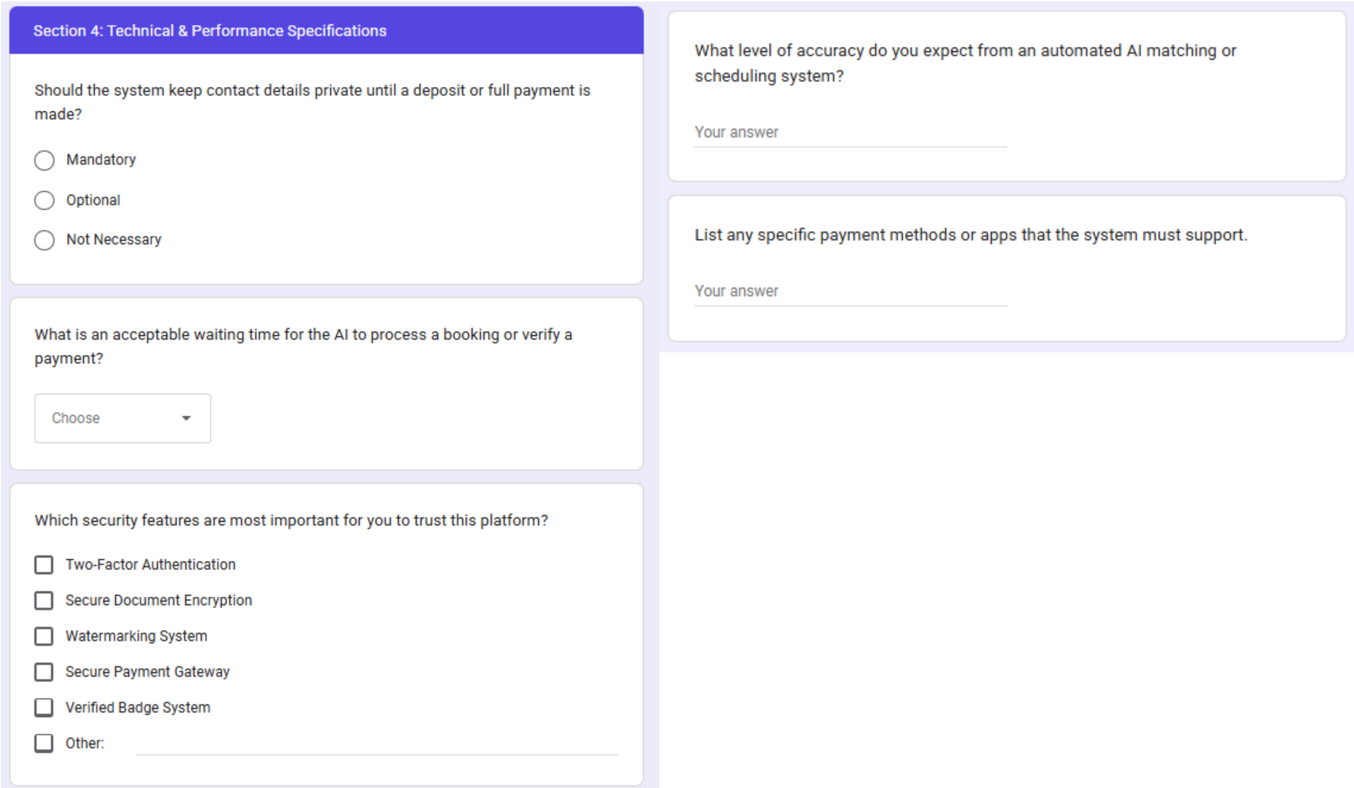
- Rescheduling
- Cancellation
- Receipt Download
- Profile Update

Describe your ideal automated notification system for booking confirmations.

Your answer

The expectation of users should always be a smooth process where you search for your preferred appointment, compare prices, review them, reserve the available slot instantly, and make payment. In case there is a need to make changes, the system should always ensure that the required time slot is checked before any changes can be allowed to avoid conflicts, and cancellation should be possible with automatic refunds.

Section 4: Technical & Performance Specifications (Non-Functional)
This section covered the stringent technical standards, data privacy considerations, and security requirements needed to create platform trust.


Section 4: Technical & Performance Specifications

Should the system keep contact details private until a deposit or full payment is made?

- Mandatory
- Optional
- Not Necessary

What is an acceptable waiting time for the AI to process a booking or verify a payment?

Choose

Which security features are most important for you to trust this platform?

- Two-Factor Authentication
- Secure Document Encryption
- Watermarking System
- Secure Payment Gateway
- Verified Badge System
- Other:

What level of accuracy do you expect from an automated AI matching or scheduling system?

Your answer

List any specific payment methods or apps that the system must support.

Your answer

The non-functional criteria are very accurate. The security framework must include Multi-Factor Authentication (MFA), Encryption of Personal Details, Verified Provider Badges, and a Payment Gateway in order to provide protection from manual bank transfer attacks. As far as performance goes, there is an upper bound for the system's response time; AI Smart Match is required to produce recommendations within 5 seconds and will have an uptime of 99%.

4.2 Summary from method used

We have gained deeper knowledge about the operations of the professional service marketplace from service seekers and freelance service experts so that we can improve on the platform we design, which is the FlexServe service discovery and booking system. Thus, the FlexServe platform can function efficiently and effectively, overcoming all the systemic bottlenecks that the current manual workflows present. For the FlexServe online marketplace, we put targeted effort into including smart automated features for the users based on the operational requirements gathered from our evaluation.

After collecting structured feedback from our questionnaire shared via Google Forms with our stakeholder, we developed a thorough understanding of freelance market constraints, current user frustrations, resource allocation boundaries, and the functional gaps of the existing AS-IS manual system. Some prominent issues with the current manual approach include fragmented tracking across multiple external applications, a lack of centralised real-time availability checks, a high risk of scheduling overlaps or double bookings, and communication delays.

We also obtained crucial ideas on specialized technical and performance criteria that must be incorporated into our proposed TO-BE System. For instance, the FlexServe online service system will be engineered to provide a comprehensive, multi-role interface enabling service provider cataloguing, interactive portfolio checking, and user-friendly navigation. Key advanced components such as an intelligent scheduling system, automated notification engine, verified provider badges, and integrated secure payment gateways will also be structurally integrated into the FlexServe ecosystem.

Besides, we utilized the Google Form questionnaire as our primary evaluation method to understand the overarching system perspectives. Since this system is being developed under strict academic and professional engineering standards, our stakeholders served as the primary representatives and evaluators of our system requirements, providing comprehensive feedback to align our proposed system modules with industry needs. Based on this expert evaluation, the stakeholder opinion is that the new system must emerge as an easier, faster, and well-managed online environment that can be seamlessly accessed by both clients and administrators to manage service orders efficiently.

5.0 Requirement Analysis

5.1 Current business process

AS-IS System (Current manual service system)

The scenarios and workflow of the current business process for the customers are as follows:

1. The individuals who are in search of service providers can find them via social media platforms like Instagram, TikTok, and Facebook.

2. When they identify the service provider whom they wish to approach, they contact them through WhatsApp or other direct messaging tools and ask them about their availability on the chosen date and time.

3. Then subsequently, they provide them with all the information related to the booking.

    3.1. This includes what kind of service they need from them, at which location, and how many people need the service.

4. If there is confirmation from the service provider about their availability, then they provide them with the quotation and the deposit payment required.

5. The client makes the deposit payments from their account into the service provider's account.

    5.1. After making the payments, they click the pictures of the receipts and send them to the service provider.

6. Subsequently, they confirm it as well and schedule the transaction into their calendar.

7. Usually, they pay the remaining part of the total payment in cash or bank transfer after getting the services done by the service provider on the particular date and time.

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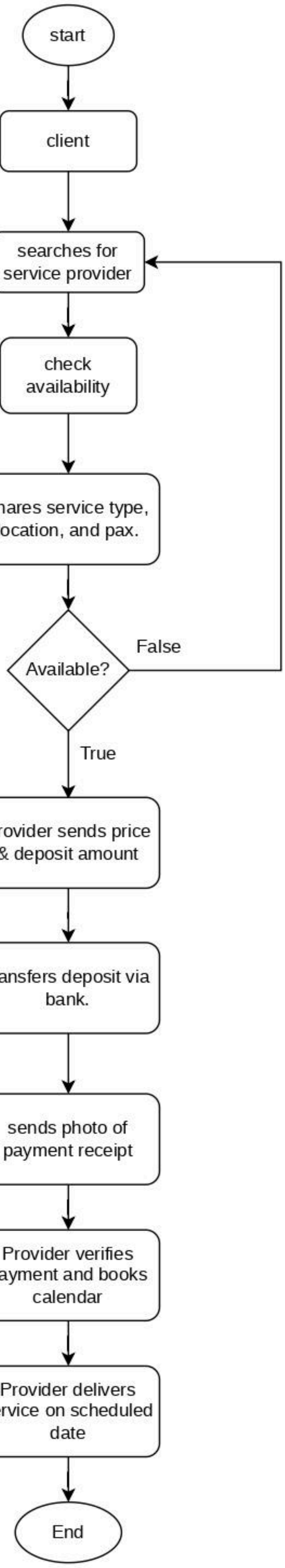

start
client
searches for service provider
check availability
shares service type, location, and pax.
Available?
False
True
Provider sends price & deposit amount
transfers deposit via bank.
sends photo of payment receipt
Provider verifies payment and books calendar
Provider delivers service on scheduled date
End

The scenarios and workflows of the current business process of the service providers are the following:

1. The service provider gets a message about the order from the client via any chosen social media channels provided in their promotional accounts: WhatsApp, Instagram, or Tik Tok.

2. The service provider confirms their availability during the required date and time.

3. The service provider gets all information about the client's wishes in order to have references in the future.

4. The service provider calculates the total price and the amount of the deposit required by the client's requests.

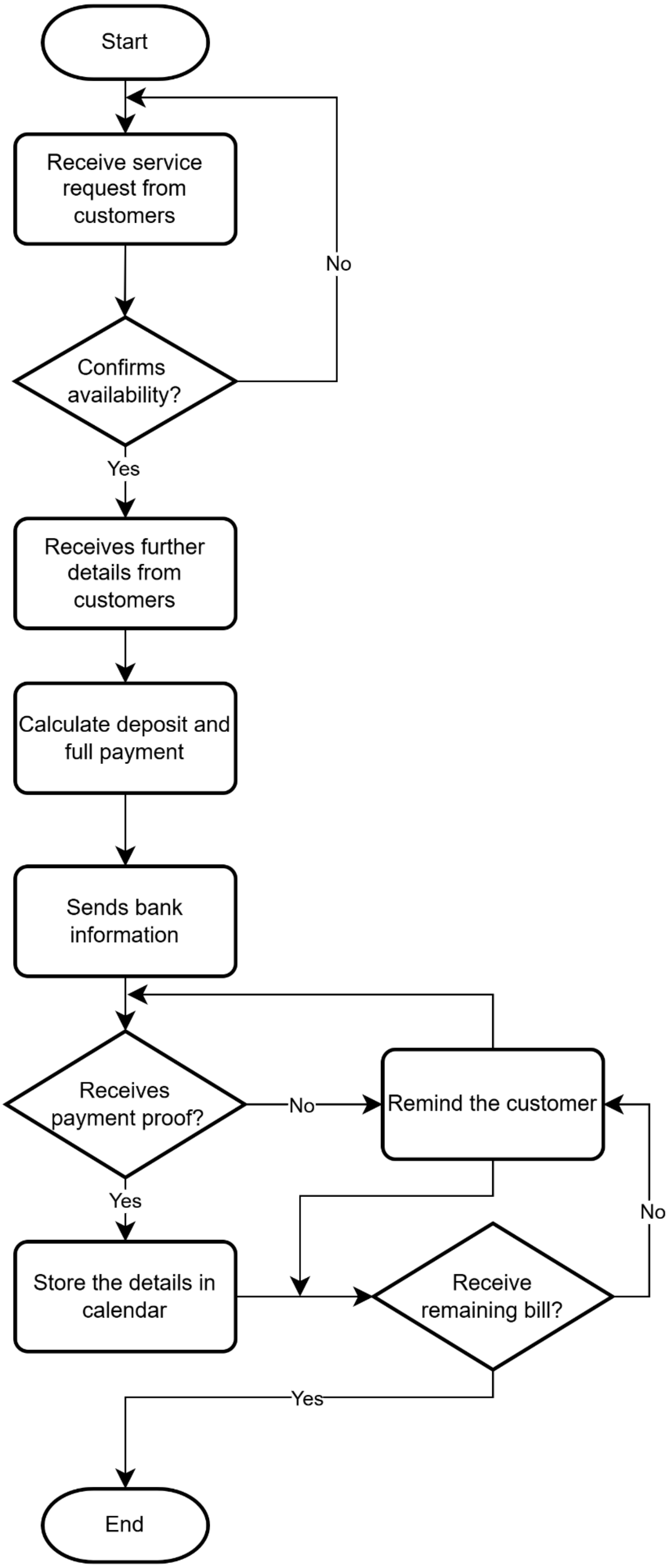
    4.1 The service provider shares their bank details to get paid the deposit from the client.

    4.2 The service provider gets a payment receipt from the client.

5. The service provider sets a reminder to themselves inside the calendar app regarding the booked date, time, place, and other necessary information.

6. The service provider sends a reminder about making the final payment.

    6.1 The service provider gets the rest of the money by the same way as the first payment.


Start
Receive service request from customers
Confirms availability?
No
Yes
Receives further details from customers
Calculate deposit and full payment
Sends bank information
Receives payment proof?
No
Remind the customer
Yes
Store the details in calendar
Receive remaining bill?
No
Yes
End

5.2 Functional Requirement

<u>AS-IS System (Current manual service system)</u>

Context diagram

Process	**Input**	**Output**
Current Service Booking System	Service request Payment proof Availability Response Payment Approval Confirmed Booking Schedule	Booking Availability Payment Receipt Booking Confirmation Booking Request Payment Notification

Level 0 diagram

Process ID	**Process**	**Input**	**Output**
1.0	Manage Booking	Service request Availability Response	Booking Availability Requirement Data Booking Request Booking details Booking Information
2.0	Process Payment	Booking Information Payment proof Payment Approval	Payment Notification Payment Receipt Payment Details Verified Payment
3.0	Confirm Booking	Verified Payment Booking Details Confirmed Booking Details	Booking Confirmation

Child Diagram

1.0 Manage Booking

Process ID	Process	Input	Output
1.1	Record Requirements	Service Request	Service Query Requirement Data
1.2	Check Availability	Service Query Availability Response	Booking Request Booking Availability Confirmed Slot
1.3	Log Booking Details	Confirmed Slot	Booking Information Booking Details

2.0 Process Payment

Process ID	Process	Input	Output
2.1	Receive Booking Information	Booking Information	Booking & Payment Information
2.2	Receive & Verify Payment	Booking & Payment Information Payment Approval Payment Proof	Payment Notification Payment Details
2.3	Store Payment Record	Payment Details	Payment Receipt Verified Payment Payment Details

3.0 Store Payment Record

Process ID	Process	Input	Output
3.1	Receive Booking Details	Booking Details	Booking Information
3.2	Verify Payment	Booking Information Verified Payments	Verified Booking Information
3.3	Prepare Booking Schedule	Verified Booking Information Confirmed Booking Schedule	Complete Booking Information
3.4	Send Booking Confirmation	Complete Booking Information	Booking Confirmation

TO-BE System (FlexServe service system)

Context Diagram

Process	Input	Output
FlexServe System	Registration Information Login Credentials Search Request Filter Preferences Booking Request Payment Information Modification Request Cancellation Request Service Information Availability Schedule Pricing Information Booking Rejection Cancellation Policy Inventory Information Buffer Recognition Payment Confirmation Transaction Status	Account status Authentication Status Search Results Recommended Service Providers Booking Confirmation Booking Status Payment Receipt Refund Status Booking Notification Booking Schedule Payment Status Revenue Analytics Report Updated Inventory Status Payment Details

Level 0 diagram

Process ID	Process	Input	Output
1.0	Account & Identity Management	Client Registration Information Provider Registration Information Client Login Credentials Provider Login Credentials Client Password Reset Request Provider Password Reset Request Client Profile Update Request Provider Profile Update Request	Client Authentication Status Provider Authentication Status Client Password Reset Approval Provider Password Reset Approval Client Profile Data Provider Profile Data Account Records
2.0	Search & AI Recommendation	Search Request Filter Preferences Listing Details	Search Results Recommended Service Providers AI Smart Match Search Activity Log
3.0	Booking & Availability	Booking Request Booking Approval Current Schedule	Availability Status Booking Schedule Booking Status Booking Records Updated Availability Schedule Booking Details Booking Total

4.0	Payment Getaway & Invoicing	Payment Information Booking Total Booking Details Transaction Status	Payment Request Payment Receipt Updated Payment Status Payment Transaction Record
5.0	Modification & Cancellation	Booking Information Modification request Cancellation request Policy Rules	Updated Booking Records e Freed Availability Slot Refund Status Refund Transaction Record
6.0	Provider Advanced Analytics Dashboard	Inventory Confirmation Buffer Configuration Revenue Request Past Revenue Data	Updated Inventory Data Updated Inventory Status Buffer Settings Confirmation Revenue Analytics Report Analytics Report

Child Diagram

1.0 Account & Identity Management

Process ID	**Process**	**Input**	**Output**
1.1	Register User	Registration Information	Account Status User Profile data

1.2	Verify Login	Login Credentials Existing User Data	Authentication Status
1.3	Manage User Profile	Profile Update Information Existing User Data	Updated Profile Status Updated User Profile

2.0 Search & AI Recommendation

Process ID	Process	Input	Output
2.1	Receive Seacrh Query	Search Request	Query Data Search Query Log
2.2	Filter Listings	Query Data Filter Preferences Provider Listings	Filtered Listings
2.3	Generate AI Recommendation	Filtered Listings Booking History	Recommended Service Providers Search Results Recommendation History

3.0 Booking & Availabilty

Process ID	Process	Input	Output
3.1	Check Availability	Availability Request Availability Schedule	Available Slots
3.2	Validate Booking	Booking Confirmation Available Slots Provider Information	Valid Booking Data Pending Booking Records

3.3	Confirm Booking	Valid Booking Data Booking Records	Booking Confirmation Booking Status Updated Booking Records

4.0 Payment Gateway & Invoicing

Process ID	Process	Input	Output
4.1	Calculate Booking Total	Payment Information Booking Records	Booking Total Payment Calculation Record
4.2	Process payment	Payment Confirmation Booking Total Payment Calculation Records	Verified Payment Transaction Status Payment Transaction Record
4.3	Generate Digital Receipt	Verified Payment Payment Transaction Record	Payment Receipt Updated Payment Status

5.0 Modification & Cancellation

Process ID	Process	Input	Output
5.1	Receive Request	Modification Request Cancellation Request Booking Request	Request Details Modification Request Record
5.2	Check Policy	Request Details Cancellation Policy	Refund Eligibility Cancellation Evaluation Record
5.3	Process Refund & Update Booking	Refund Eligibility	Refund Status

		Payment Transaction Record	Updated Booking Status Updated Booking Record

6.0 Provider Advanced Analytics Dashboard

Process ID	Process	Input	Output
6.1	Upload Inventory	Inventory Information Provider Information	Updated Inventory status Updated Inventory Data
6.2	Configure Buffer Parameters	Buffer Configuration Booking Records	Buffer Settings Confirmation Updated Buffer Settings
6.3	Generate Revenue Analytics	Revenue Request Payment Records	Revenue Analytics Report Analytics Report Data

Functional Requirement (To-Be System)

Process ID	**Functional Requirement**
1.0	The system should permit a user to create an account as either a service provider or as a client. The system should authenticate a user using a secure login process. The system should enable service providers to create and maintain their profiles that include services offered, portfolio, and availability.
2.0	The system will enable users to look for service providers. The system will enable users to filter providers by categories, cost, ratings, and availability. The system will recommend service providers according to user preference and booking history.
3.0	The system should validate availability and ensure there are no booking conflicts. The system should support online payment processing and maintain transaction records. The system should automatically generate and send payment receipts via email upon completion of a transaction.
4.0	PThe application should show available booking slots. The application should validate the booking request. The application should confirm successful bookings.
5.0	The system should enable users to modify their bookings. The system should enable users to cancel their bookings based on policy rules. The system should handle refunds on cancellations of bookings based on rules.

6.0	The system should enable providers to manage inventory. The system should enable providers to set booking buffer times. The system should provide revenue analytics reports.

5.3 Non-functional Requirement

For ensuring smooth, secure, and reliable operation of the FlexServe service infrastructure, the system needs to meet a particular set of non-functional requirements. These performance standards are further classified into performance objectives and control standards to provide not only a great user experience but also high operational standards.

Performance Requirements

System Availability:

- The system should have a 99.9% uptime ratio so that its availability is ensured for both customers and service providers, especially during periods of peak activity when bookings take place.

Response Time:

- In order to ensure the smooth operation of the user interaction, the system must respond to search requests and service requests within 10 seconds for at least 95% of transactions.

Scalability:

- The scaling process needs to be done dynamically for coping with traffic peaks, allowing up to 15,000 online users simultaneously while maintaining the same level of performance.

Data Processing:

- It is important for the real-time update to occur; therefore, any updates on provider schedule, status of service booking, or completion of payments should take place in 1 second after submission.

Throughput:

- Both the database system and the payment gateways must have the capacity to process a minimum of 600 transactions per second in order to handle the peak workloads efficiently (during flash sales or seasonal campaigns).

Backup and Recovery:

- There should be daily automated backup of all transaction information and systems data. The recovery time after an infrastructure disaster should not exceed 4 hours.

Compatibility:

- This interface should provide users with a seamless experience on all modern web browsers (Chrome, Safari, Firefox, Edge) and mobile platforms (iOS and Android).

Resource Utilization:

- In order to avoid any unforeseen crashes of the system, the resource management on the back-end should be maximized so that the CPU and memory usage on servers remain below 75%.

Control Requirements

Security:

- **Data Encryption:** All sensitive information, including user profiles, identification records, and payment credentials, must be protected using AES-256 encryption both when stored at rest and while in transit.
- **Authentication:** Multi-factor authentication (MFA) must be enforced for administrators and service providers to safeguard financial dashboards and prevent unauthorized profile access.
- **Access Control:** A strict Role-Based Access Control (RBAC) mechanism must separate privileges clearly between Customers, Service Providers, and FlexServe System Administrators.

Data Integrity:

- **Validation:** The user inputted values, such as hourly pay, date of reservation, and telephone number, must go through strict client-side and server-side validation for security purposes so as to avoid injection attacks and data damage.
- **Audit Trails:** The system is to provide unalterable audit logs recording all major user actions such as registration, booking changes, cancellations, and payout disbursements.

Compliance:

- **Regulatory Compliance:** To safeguard user privacy, data handling and retention policies must comply fully with relevant regional data protection frameworks (such as PDPA or GDPR).
- **Standards Adherence:** System development must align with recognized web application security standards, specifically addressing the vulnerabilities outlined in the OWASP Top Ten.

Reliability:

- **Error Handling:** A system that has a fault-tolerance architecture that can record and log all technical faults on the back-end while showing meaningful error messages to the users but not the actual system codes is necessary.
- **Redundancy:** Redundancy should always be set up in cloud infrastructure, specifically for mission-critical elements within applications, matches engines, and core databases in case of any failure of a computer.

Usability:

- **User Interface:** The platform layout must remain clean, highly intuitive, and easily navigable, ensuring accessibility for individuals with varying levels of technical literacy.
- **Training and Documentation:** Clear in-app onboarding flows, interactive FAQs, and structured documentation must be available to help new service providers set up and manage their professional profiles seamlessly.

Maintainability:

- **Modular Design:** The system should follow a modular or microservices-based architecture, allowing the development team to roll out feature updates, conduct maintenance, or scale individual modules independently.
- **Monitoring:** Continuous automated monitoring tools must actively track backend performance, log query delays, and trigger immediate alerts for the engineering team if any operational anomalies occur.

Privacy:

- **Data Minimization:** To reduce exposure risks, the platform must limit its data collection to information that is strictly necessary for service matching, profile verification, and payment processing.
- **User Consent:** The system must request explicit permission from users before accessing device location tracking (used for locating nearby service providers) or sending push notifications, offering simple ways to opt out at any time.

AI Perfomance Metrics

Response Time:

- The AI Smart Match module will produce recommendations within 5 seconds following the submission of a search request by a user.

Recommendation Accuracy:

- The AI Smart Match module will make recommendations which are 85% relevant to user preferences, search history, and booking history.

Recommendation Availability:

- The AI Smart Match service will be available 99% of times when providing recommendations.

Data Freshness:

- The AI Smart Match module will rely on the latest data related to users' profile, booking history, and providers' information.

Payment Integrity

Transaction Verification:

- All payment transactions must undergo verification prior to issuance of confirmation of the transaction.

Receipt Generation:

- A receipt will be automatically generated by the system for each completed payment transaction.

Transaction Logging:

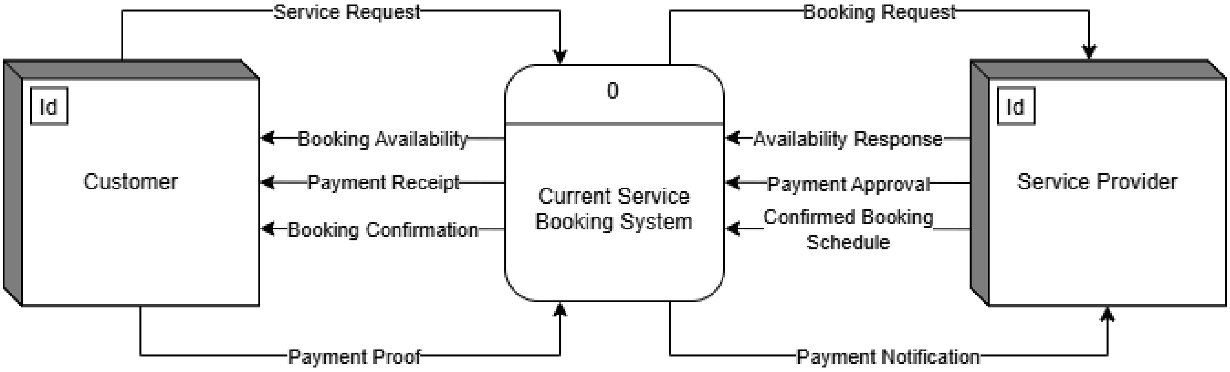
- All transactions will be properly logged into the system for audit purposes.

Duplicate Payment Prevention:

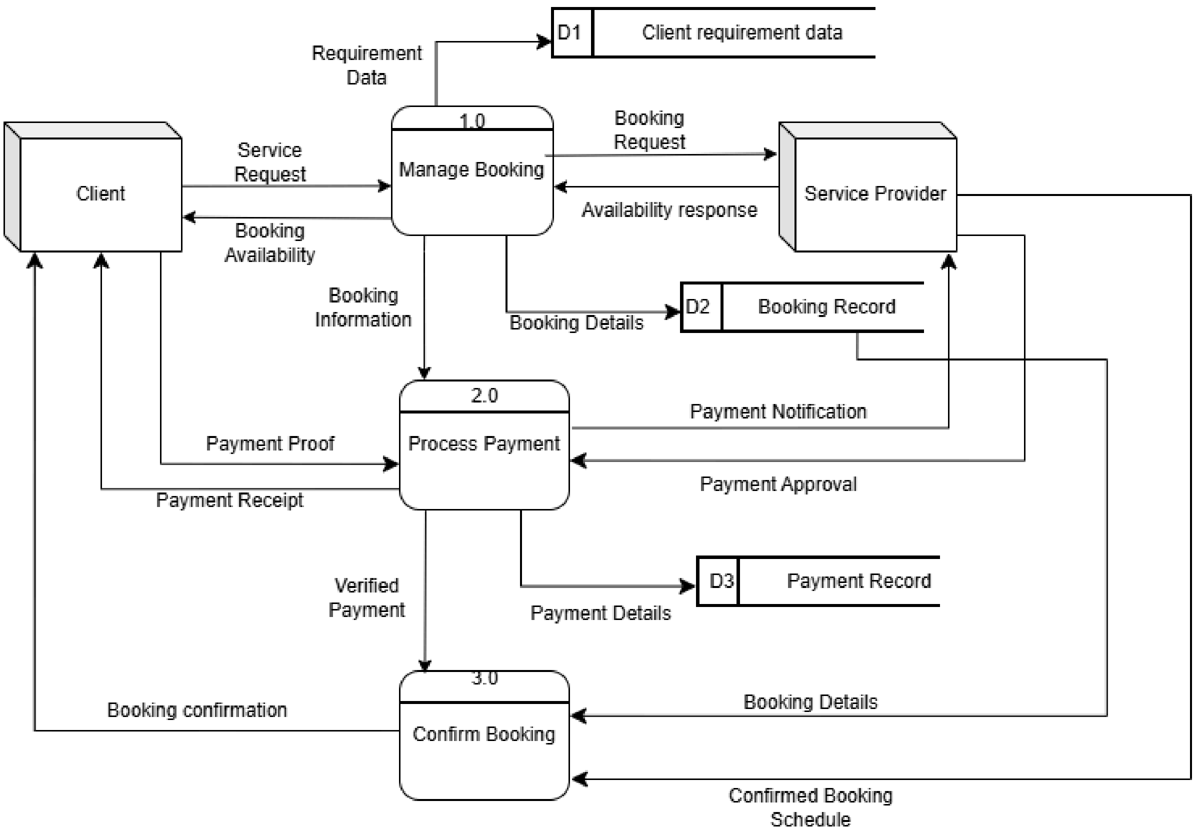
- Duplicate payment submissions must be prevented.

5.4 Logical DFD AS-IS system

5.4.1 Context Diagram

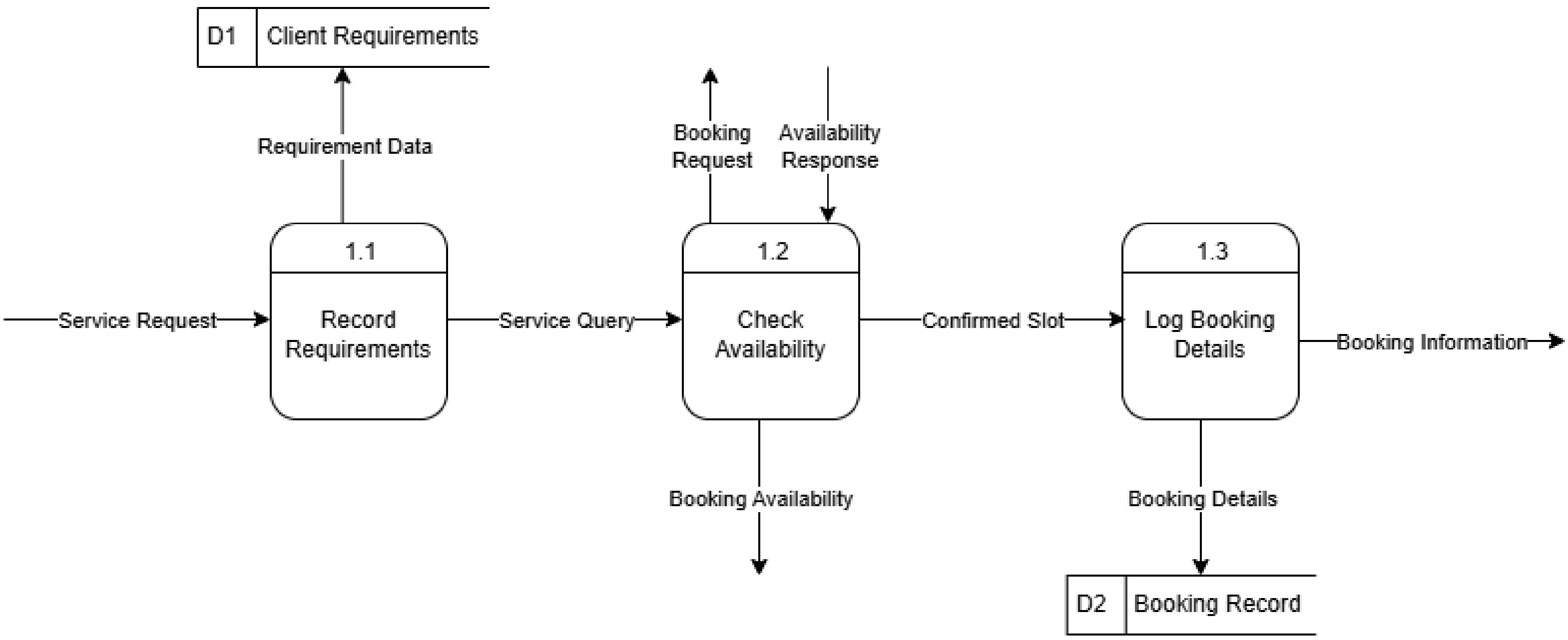


5.4.2 Diagram 0

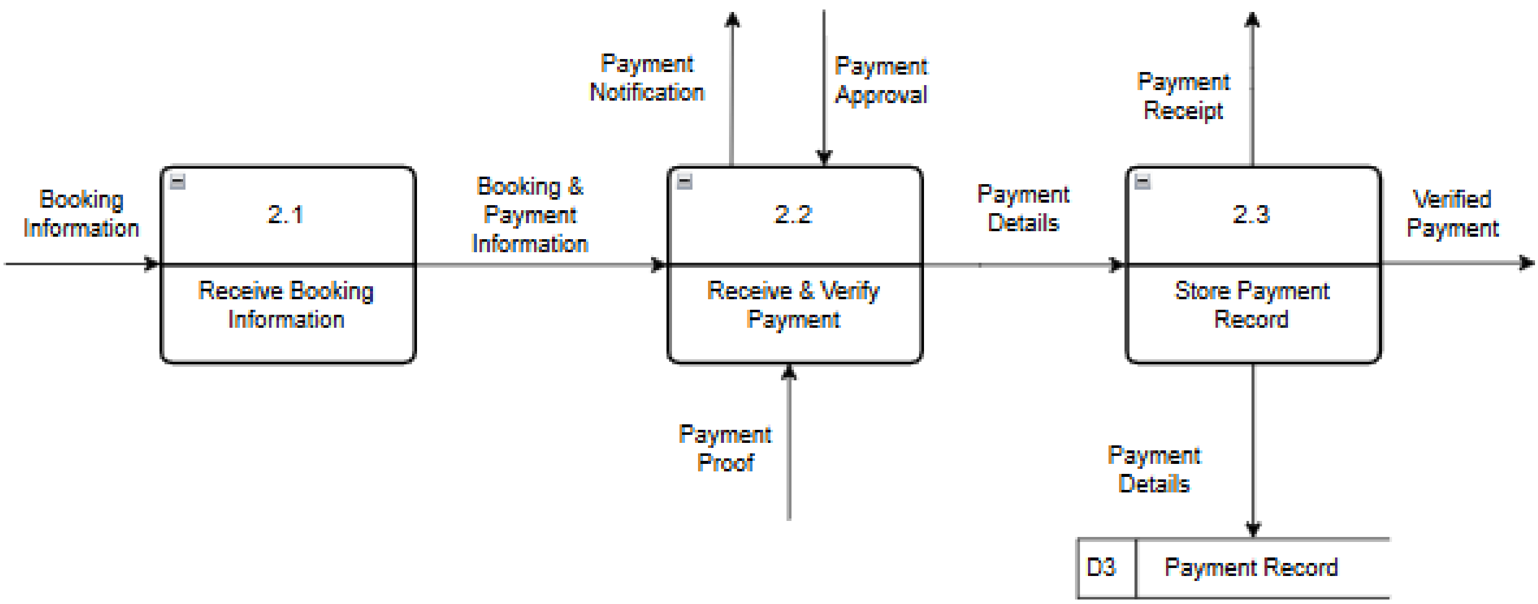


5.4.3 Child Diagram

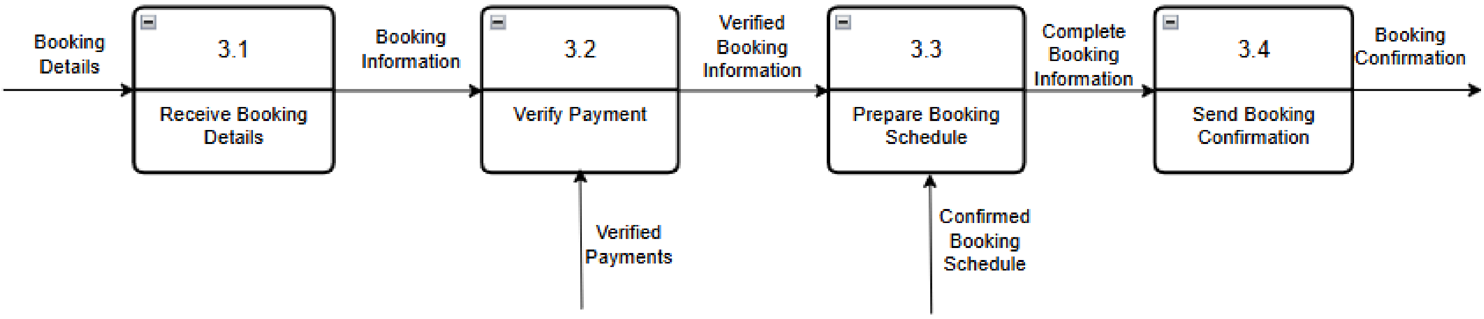
1.0 Manage Booking



2.0 Process Payment



3.0 Store Payment Record



6.0 Summary of Requirement Analysis process

Flexserve is designed to make the interaction and transaction between service providers and seekers to be seamless and secure. The customers would be able to find qualified service providers through Flexserve platform thanks to its sorting system. Meanwhile, service providers can do business with verified customers which promises serious and legit proposals through its registration requirement feature.

Flexserve compiles all the required functions of a service provider and a customer need into one single platform. This will significantly reduce the redundancy in data and overlapped schedules for the service provider and reduce time taken for customers to find the preferred service without having to search into multiple social media platforms.

Furthermore, Flexserve offers security from secure data management to prevent the user's private data from being exploited by any third-party. The FlexServe AI Smart Match Technology helps in recommending the best service providers to the customer according to his/her preferences and the booking requirements of the service..

Based on the analysis conducted, Flexserve is very capable in meeting user’s expectations and requirements while improving operational efficiency and accessibility. Therefore the development of the platform is a practical and beneficial solution for the service-for-hire market.